

Hollow Core Photonic Crystal Fibre: Perfect Light Pipe?

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In 1991 I proposed a new kind of optical fibre, in which light is guided by photonic band gap (PBG) effects, side-stepping the need for total internal reflection and potentially allowing low-loss guidance in a hollow core [1–4]. Over the intervening decades this idea has evolved and our understanding of the guidance mechanisms, and how they can be optimised, has made remarkable progress (indeed, the latest hollow-core fibres do not have a PBG, but can be viewed as guiding light instead by a radial stop-band). As a result, the transmission loss has steadily fallen, making the latest single-ring hollow-core PCFs (or “anti-resonant hollow core fibres”) realistic challengers to conventional all-solid step-index fibre, while offering a much wider usable transmission window, and much lower Rayleigh scattering and optical nonlinearity. The scene seems set for the adoption of hollow-core fibres in long-distance data transmission at wavelengths beyond the 1550 nm band, perhaps even in the near IR.

The quest for a low-loss hollow-core optical fibre (HCF) goes back to the 1960s, starting with a much-cited paper by Marcatili and Schmeltzer who explored theoretically the possibility of using simple glass capillaries for light guidance in optical masers, but concluded that the loss was too high for use in optical telecommunications [5]. In the 2000s, Harrington worked on using capillaries, both coated and uncoated on the inner surface with highly reflective materials, for short-range high power transmission and sensing in the infrared [6,7]. Confinement by Bragg reflection at a cylindrical multilayer mirror, first suggested in the 1970s by Yeh, Yariv and Marom [8,9], has been used more recently to form HCFs for 10 μm light, with applications in eye surgery [10]. However, none of these fibres has delivered transmission loss low enough for use in long-distance optical communications. There seemed, however, a good chance that a microstructured fibre, guiding by photonic bandgap (PBG) effects and made only from pure fused silica, might deliver very low transmission loss, perhaps even below that of conventional step-index fibre [4].

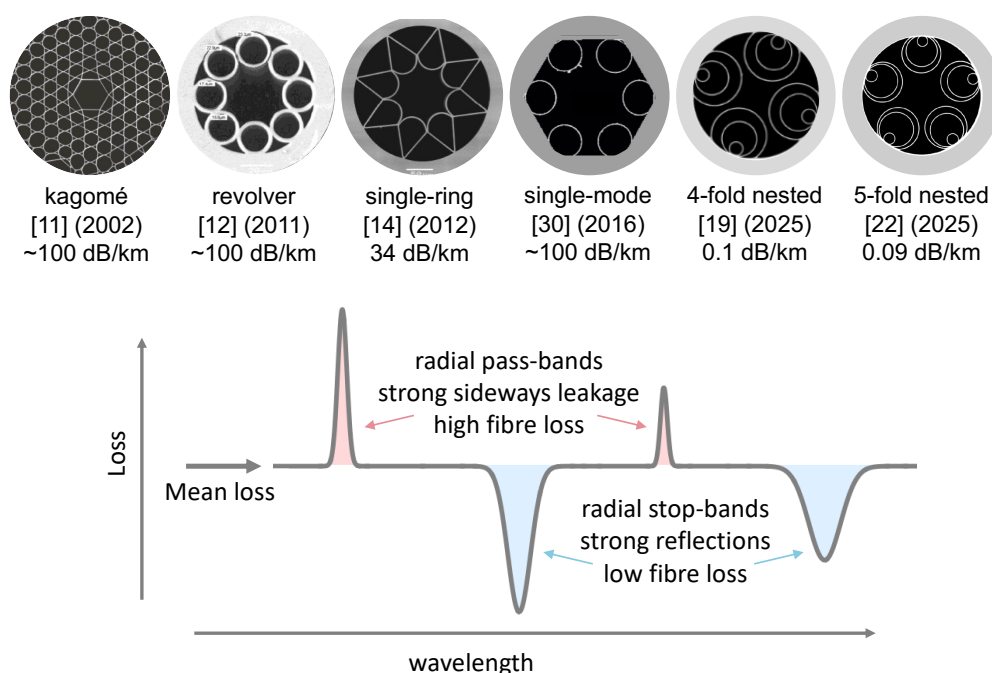


Fig. 1: Upper: Micrographs of the structure of five different single-ring PCFs. The lowest loss values (last two on the right) were measured at 1.55 μm . Lower: Spectral transmission loss of the fundamental core mode of a typical single-ring PCF (conceptual only). The capillary ring microstructure creates radial stop- and pass-bands, yielding narrow bands of low and high loss at a series of different wavelengths.

The first hollow core PBG-guiding fibre, showing a series of spectrally narrow transmission bands superimposed on a high-loss background, was reported in 1999 [2]. In 2003 the company Blazephotonics, a pioneering spin-off from the University of Bath, succeeded in reducing the 1550 nm loss of these fibres to 0.8 dB/km, but ran out of funds and was forced to close down. Around the same time the Bath group also reported a

kagomé-lattice fibre delivering broad-band guidance with moderately low (below ~ 1 dB/m) loss [11]. Numerical analysis of this fibre showed that its cladding did not support a full photonic bandgap, indicating that some other guidance mechanism was at work. In 2011 a team in Moscow reported a “single-ring” PCF (SR-PCF), consisting of a revolver-like ring of touching capillaries mounted inside a larger thick-walled cladding tube, resembling a kagomé-fibre with only the first row of hollow channels retained, and blown up into circular capillaries [12,13], and delivering narrow bands of low loss in the infrared. This triggered world-wide interest in SR-PCFs with ever more complex “nested” cladding structures and steadily decreasing transmission loss at wavelengths from the infrared to the UV [14–21]. The most recent of these fibres shows loss at 1550 nm below that of the best telecommunications fibres [19,22]. So far, however, the most important applications of these fibres have been in fields such as gas-based ultrafast nonlinear optics [23–25], delivery of high power CW and pulsed laser light for industrial machining [20], particle trapping and propulsion [26], and efficient quantum frequency conversion of single photons [27]. There has already been commercial success outside telecommunications: a recent ASML lithography machine uses gas-filled SR-PCF to generate bright broad-band UV light [28].

SR-PCFs typically exhibit a series of spectrally narrow high-loss and low-loss bands superimposed on a broad-band moderately low-loss background (Fig. 1). Their behaviour is dominated by the photonic band structure of the single-ring cladding, which may be usefully viewed as having radial stop- and pass-bands at different wavelengths that act upon the fundamental core mode, either creating high loss (at a pass-band), or low loss when a radial stop-band forms in the capillary structure. These effects are particularly strong very close to glancing incidence when air-glass Fresnel reflections are very high and the transverse effective wavelength of the core light is much greater than any of the fine structural features of the cladding ring, which cannot be “imaged” by the core light, trapping it strongly in the core. By ensuring that the core light is anti-resonant with the glass walls of the capillaries, the light-glass overlap can be made very small ($< 0.1\%$), further reducing the loss and the scattering.

The most recent nested SR-PCFs can thus be viewed as guiding by a modified form of cylindrical Bragg reflection, as proposed in 1976 [8]. Curiously, this is reminiscent of the first solid core step-index PCFs that guided by a modified form of total internal reflection [29]. Both these fibres have an azimuthally periodic structure that confers unique additional features, in particular modal filtering [30] and twist-sensitivity [31].

In the tutorial I plan to briefly review the history of hollow-core PCF, explore its many applications, and outline how “glancing incidence optics” has revolutionised hollow-core fibre design.

References

- [1] P. St.J. Russell, T. A. Birks, and F. D. Lloyd-Lucas, *Photonic Bloch Waves and Photonic Bandgaps*, in *NATO Advanced Study Institute (ASI) on Confined Electrons and Photons*, Vol. NSSB 340 (Erice, Italy, 1993), pp. 585–633.
- [2] R. F. Cregan, B. J. Mangan, J. C. Knight, T. A. Birks, P. St.J. Russell, P. J. Roberts, and D. C. Allan, Single-mode photonic band gap guidance of light in air, *Science* **285**, 1537 (1999).
- [3] P. St.J. Russell, Photonic crystal fibers, *Science* **299**, 358 (2003).
- [4] P. Russell, Invited commentary: hollow core glass fibre—the perfect pipe for light, *eLight* **5**, 4 (2025).
- [5] E. A. J. Marcatili and R. A. Schmeltzer, Hollow metallic and dielectric waveguides for long distance optical transmission and lasers, *Bell Syst. Tech. J.* **43**, 1783 (1964).
- [6] J. A. Harrington, A review of IR transmitting, hollow waveguides, *Fiber and Integrated Optics* **19**, 211 (2000).
- [7] R. George and J. A. Harrington, Cu/CuI-coated hollow glass waveguides for delivery of infrared radiation, *Optical Engineering* **45**, (2006).
- [8] P. Yeh, A. Yariv, and E. Marom, Theory of Bragg fiber, *J. Opt. Soc. Am.* **68**, 1196 (1978).
- [9] J. L. Archambault, R. J. Black, S. Lacroix, and J. Bures, Loss calculations for antiresonant wave-guides, *J. Lightwave Tech.* **11**, 416 (1993).
- [10] B. Temelkuran, S. D. Hart, G. Benoit, J. D. Joannopoulos, and Y. Fink, Wavelength-scalable hollow optical fibres with large photonic bandgaps for CO(2) laser transmission, *Nature* **420**, 650 (2002).
- [11] F. Benabid, J. C. Knight, G. Antonopoulos, and P. St.J. Russell, Stimulated Raman scattering in hydrogen-filled hollow-core photonic crystal fiber, *Science* **298**, 399 (2002).
- [12] A. D. Pryamikov, A. S. Biriukov, A. F. Kosolapov, V. G. Plotnichenko, S. L. Semjonov, and E. M. Dianov, Demonstration of a waveguide regime for a silica hollow-core microstructured optical fiber with a negative curvature of the core boundary in the spectral region $< 3.5 \mu\text{m}$, *Opt. Exp.* **19**, 1441 (2011).
- [13] B. Debord, M. Alharbi, T. Bradley, C. Fourcade-Dutin, Y. Y. Wang, L. Vincetti, F. Gérôme, and F. Benabid, Hypocycloid-shaped hollow-core photonic crystal fiber Part I: Arc curvature effect on confinement loss, *Opt. Exp.* **21**, 28597 (2013).
- [14] F. Yu, W. J. Wadsworth, and J. C. Knight, Low loss silica hollow core fibers for 3–4 micrometre spectral region, *Opt. Exp.* **20**, 11153 (2012).
- [15] B. Debord et al., Ultralow transmission loss in inhibited-coupling guiding hollow fibers, *Optica* **4**, 209 (2017).
- [16] S. Gao, Y. Wang, W. Ding, D. Jiang, S. Gu, X. Zhang, and P. Wang, Hollow-core conjoined-tube negative-curvature fibre with ultralow loss, *Nat. Comm.* **9**, 2828 (2018).
- [17] E. N. Fokoua, S. Abokhamis Mousavi, G. T. Jasion, D. J. Richardson, and F. Poletti, Loss in hollow-core optical fibers: mechanisms, scaling rules, and limits, *Adv. Opt. Photon.* **15**, 1 (2023).
- [18] R. Mears, K. Harrington, W. J. Wadsworth, J. C. Knight, J. M. Stone, and T. A. Birks, Guidance of ultraviolet light down to 190 nm in a hollow-core optical fibre, *Opt. Express* **32**, 8520 (2024).
- [19] S. Gao, H. Chen, Y. Sun, Y. Xiong, Z. Yang, R. Zhao, W. Ding, and Y. Wang, Fourfold truncated double-nested antiresonant hollow-core fiber with ultralow loss and ultrahigh mode purity, *Optica* **12**, 56 (2025).
- [20] W. Lin et al., Flexible delivery of broadband, 100 fs mid-infrared pulses in the water-absorption band using hollow-core photonic crystal fiber, *Optica* **12**, 901 (2025).
- [21] C. Wei, R. Joseph Weiblen, C. R. Menyuk, and J. Hu, Negative curvature fibers, *Advances in Optics and Photonics* **9**, 504 (2017).
- [22] M. Petrovich et al., Broadband optical fibre with an attenuation lower than 0.1 decibel per kilometre, *Nat. Photon.* **19**, 1203 (2025).
- [23] J. C. Travers, W. Chang, J. Nold, N. Y. Joly, and P. St.J. Russell, Ultrafast nonlinear optics in gas-filled hollow-core photonic crystal fibers, *J. Opt. Soc. Am. B* **28**, A11 (2011).

- [24] P. St.J. Russell, P. Hölzer, W. Chang, A. Abdolvand, and J. C. Travers, Hollow-core photonic crystal fibres for gas-based nonlinear optics, *Nat. Phot.* **8**, 278 (2014).
- [25] U. Elu et al., Seven-octave high-brightness and carrier-envelope-phase-stable light source, *Nature Photonics* **15**, 277 (2021).
- [26] M. N. Romodina, S. Xie, F. Tani, and P. St.J. Russell, Backward jet propulsion of particles by femtosecond pulses in hollow-core photonic crystal fiber, *Optica* **9**, 268 (2022).
- [27] R. Tyumenev, J. Hammer, N. Y. Joly, P. St.J. Russell, and D. Novoa, Tunable and state-preserving frequency conversion of single photons in hydrogen, *Science* **376**, 621 (2022).
- [28] www.asml.com/products/metrology-and-inspection-systems/yeildstar-1390
- [29] T. A. Birks, J. C. Knight, and P. St.J. Russell, Endlessly single-mode photonic crystal fiber, *Opt. Lett.* **22**, 961 (1997).
- [30] P. Uebel, M. C. Günendi, M. H. Frosz, G. Ahmed, N. N. Edavalath, J.-M. Ménard, and P. St.J. Russell, Broadband robustly single-mode hollow-core PCF by resonant filtering of higher-order modes, *Opt. Lett.* **41**, 1961 (2016).
- [31] P. Roth, Y. Chen, M. Günendi, R. Beravat, N. Edavalath, M. Frosz, G. Ahmed, G. Wong, and P. St.J. Russell, Strong circular dichroism for the HE₁₁ mode in twisted single-ring hollow-core photonic crystal fiber: Erratum, *Optica* **9**, 1037 (2022).